

PMSCS 694

Information Security

Crypto: Part V

Lecture 7

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Public Key Cryptography

- ❑ Two keys, one to encrypt, another to decrypt
 - Alice uses Bob's **public key** to encrypt
 - Only Bob's **private key** decrypts the message
- ❑ Based on "trap door, one way function"
 - "One way" means easy to compute in one direction, but hard to compute in other direction
 - Example: Given p and q , product $N = pq$ easy to compute, but hard to find p and q from N
 - "Trap door" is used when creating key pairs

Public Key Cryptography

□ Encryption

- Suppose we **encrypt** M with Bob's public key
- Bob's private key can **decrypt** C to recover M

□ Digital Signature

- Bob **signs** by "encrypting" with his private key
- Anyone can **verify** signature by "decrypting" with Bob's public key
- But only Bob could have signed
- Like a handwritten signature, but much better...

Modular Inverse

Example 3: Find the M.I of 11 mod 26.

Q	A	B	R	T ₁	T ₂	T
2	26	11	4			

11 $\overline{) \begin{array}{r} 26 \\ 22 \\ \hline 4 \end{array}}$

Example 3: Find the M.I of 11 mod 26.

Q	A	B	R	T_1	T_2	T
2	26	11	4	0	1	-2

$$T_1 = 0 \text{ and } T_2 = 1$$

$$T = T_1 - T_2 \times Q$$

$$T = 0 - 1 \times 2$$

$$T = 0 - 2$$

$$T = -2$$

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2	26	11	4	0	1	-2
	11	4		1	-2	

Example 3: Find the M.I of 11 mod 26.

Q	A	B	R	T ₁	T ₂	T
2	26	11	4	0	1	-2
2	11	4		1	-2	

$$\begin{array}{r}
 2 \\
 \hline
 4 \overline{) 11} \\
 \underline{8} \\
 3
 \end{array}$$

Example 3: Find the M.I of 11 mod 26.

Q	A	B	R	T ₁	T ₂	T
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5

$$T = T_1 - T_2 \times Q$$

$$T = 1 - (-2) \times 2$$

$$T = 1 - (-4)$$

$$T = 1 + 4$$

$$T = 5$$

Example 3: Find the M.I of 11 mod 26.

Q	A	B	R	T ₁	T ₂	T
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
3	3	1	0	5	-7	26
X	1	0	X	-7	26	X

0 1



Example 3: Find the M.I of 11 mod 26.

Q	A	B	R	T ₁	T ₂	T
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
3	3	1	0	5	-7	26
X	1	0	X	19	26	X

$-7 \bmod 26 = -7 + 26 = 19$

$\therefore 19$ is the M.I of 11 mod 26.

Knapsack



Knapsack Problem

- Given a set of n weights $W_0, W_1, \dots, W_{n-1}$ and a sum S , find $a_i \in \{0, 1\}$ so that

$$S = a_0 W_0 + a_1 W_1 + \dots + a_{n-1} W_{n-1}$$

(technically, this is the *subset sum* problem)

- **Example**

- Weights (62, 93, 26, 52, 166, 48, 91, 141)
- Problem: Find a subset that sums to $S = 302$
- Answer: $62 + 26 + 166 + 48 = 302$

- The (general) knapsack is NP-complete

Knapsack Problem

- ❑ **General knapsack (GK)** is hard to solve
- ❑ But **superincreasing knapsack (SIK)** is easy
- ❑ SIK —each weight greater than the *sum of all previous weights*
- ❑ **Example**
 - Weights (2,3,7,14,30,57,120,251)
 - Problem: Find subset that sums to $S = 186$
 - Work from largest to smallest weight
 - Answer: $120 + 57 + 7 + 2 = 186$

Knapsack Cryptosystem

1. Generate superincreasing knapsack (SIK)
 2. Convert SIK to "general" knapsack (GK)
 3. **Public Key:** "general" knapsack
 4. **Private Key:** SIK and **conversion factor**
- **Goal...**
- Easy to encrypt with GK
 - With private key, easy to decrypt (solve SIK)
 - Without private key, Trudy has no choice but to try to solve GK

Example

- Start with (2,3,7,14,30,57,120,251) as the SIK
- Choose $m = 41$ and $n = 491$ (m, n relatively prime, n exceeds sum of elements in SIK)
- Compute "general" knapsack
 - $2 \cdot m \bmod 491 = 2 \cdot 41 \bmod 491 = 82$
 - $3 \cdot m \bmod 491 = 3 \cdot 41 \bmod 491 = 123$
 - $7 \cdot m \bmod 491 = 7 \cdot 41 \bmod 491 = 287$
 - $14 \cdot m \bmod 491 = 14 \cdot 41 \bmod 491 = 83$
 - $30 \cdot m \bmod 491 = 30 \cdot 41 \bmod 491 = 248$
 - $57 \cdot m \bmod 491 = 57 \cdot 41 \bmod 491 = 373$
 - $120 \cdot m \bmod 491 = 120 \cdot 41 \bmod 491 = 10$
 - $251 \cdot m \bmod 491 = 251 \cdot 41 \bmod 491 = 471$
- "General" knapsack: (82,123,287,83,248,373,10,471)

Knapsack Example

□ **Private key:** (2,3,7,14,30,57,120,251)

$$m^{-1} \bmod n = 41^{-1} \bmod 491 = 12$$

□ **Public key:** (82,123,287,83,248,373,10,471), $n=491$

□ **Example: Encrypt $M=150$, which is 10010110**

$$82 + 83 + 373 + 10 = 548$$

□ **To decrypt, use private key...**

- $548 \cdot 12 = 193 \bmod 491$

- Solve (easy) SIK with $S = 193$

- Obtain plaintext 10010110

Knapsack Proof

- Example: Encrypt $M=150$, which is 10010110

$$82 + 83 + 373 + 10 = 548$$

- To decrypt, use private key...

- $548 \cdot m^{-1} = 82 \cdot m^{-1} + 83 \cdot m^{-1} + 373 \cdot m^{-1} + 10 \cdot m^{-1}$
- $= 2mm^{-1} + 14mm^{-1} + 57mm^{-1} + 120mm^{-1}$
- $= 2 + 14 + 57 + 120$
- $= 193 \bmod 491$

Knapsack Weakness

- ❑ **Trapdoor:** Convert SIK into "general" knapsack using modular arithmetic
- ❑ **One-way:** General knapsack easy to encrypt, hard to solve; SIK easy to solve
- ❑ This knapsack cryptosystem is **insecure**
 - Broken in 1983 with Apple II computer
 - The attack uses **lattice reduction**
- ❑ "General knapsack" is not general enough!
 - This special case of knapsack is easy to break

RSA

RSA

- ❑ Invented by Clifford Cocks (GCHQ) and Rivest, Shamir, and Adleman (MIT)
 - RSA is the *gold standard* in public key crypto
- ❑ Let p and q be two large prime numbers
- ❑ Let $N = pq$ be the **modulus**
- ❑ Choose e relatively prime to $(p-1)(q-1)$
- ❑ Find d such that $ed = 1 \bmod (p-1)(q-1)$
- ❑ **Public key** is (N, e)
- ❑ **Private key** is d

RSA

- Message M is treated as a number
- To encrypt M we compute
$$C = M^e \bmod N$$
- To decrypt ciphertext C , we compute
$$M = C^d \bmod N$$
- Recall that e and N are public
- If Trudy can factor $N = pq$, she can use e to easily find d since $ed = 1 \bmod (p-1)(q-1)$
- So, **factoring the modulus breaks RSA**
 - Is factoring the only way to break RSA?

Does RSA Really Work?

- Given $C = M^e \bmod N$ we want to show that $M = C^d \bmod N = M^{ed} \bmod N$
- We'll need **Euler's Theorem**:
If x is relatively prime to n then $x^{\phi(n)} = 1 \bmod n$
- **Facts**:
 - 1) $ed = 1 \bmod (p-1)(q-1)$
 - 2) By definition of "mod", $ed = k(p-1)(q-1) + 1$
 - 3) $\phi(N) = (p-1)(q-1)$
- Then $ed - 1 = k(p-1)(q-1) = k\phi(N)$
- So, $C^d = M^{ed} = M^{(ed-1)+1} = M \cdot M^{ed-1} = M \cdot M^{k\phi(N)}$
 $= M \cdot (M^{\phi(N)})^k \bmod N = M \cdot 1^k \bmod N = \mathbf{M \bmod N}$

RSA Keypair

- ❑ Generate RSA key pair...
 - Select “large” primes $p = 11, q = 3$
 - Then $N = pq = 33$ and $(p - 1)(q - 1) = 20$
 - Choose $e = 3$ (relatively prime to 20)
 - Find d such that $ed = 1 \pmod{20}$
 - We find that $d = 7$ works
- ❑ **Public key:** $(N, e) = (33, 3)$
- ❑ **Private key:** $d = 7$

Textbook RSA Example

- ❑ **Public key:** $(N, e) = (33, 3)$
- ❑ **Private key:** $d = 7$
- ❑ Suppose message to encrypt is $M = 8$
- ❑ Ciphertext C is computed as
$$C = M^e \bmod N = 8^3 = 512 = 17 \bmod 33$$
- ❑ Decrypt C to recover the message M by
$$\begin{aligned} M &= C^d \bmod N = 17^7 = 410,338,673 \\ &12,434,505 * 33 + 8 = 8 \bmod 33 \end{aligned}$$
- ❑ Why is this "textbook" RSA?

More Efficient RSA (1)

❑ Modular exponentiation example

- $5^{20} = 95367431640625 = 25 \pmod{35}$

❑ A better way: repeated squaring

- $20 = 10100$ base 2
- $(1, 10, 101, 1010, 10100) = (1, 2, 5, 10, 20)$
- Note that $2 = 1 \cdot 2$, $5 = 2 \cdot 2 + 1$, $10 = 2 \cdot 5$, $20 = 2 \cdot 10$
- $5^1 = 5 \pmod{35}$
- $5^2 = (5^1)^2 = 5^2 = 25 \pmod{35}$
- $5^5 = (5^2)^2 \cdot 5^1 = 25^2 \cdot 5 = 3125 = 10 \pmod{35}$
- $5^{10} = (5^5)^2 = 10^2 = 100 = 30 \pmod{35}$
- $5^{20} = (5^{10})^2 = 30^2 = 900 = 25 \pmod{35}$

❑ No huge numbers and it's efficient!

More Efficient RSA (2)

- ❑ Use $e = 3$ for all users (but not same N or d)
 - + Public key operations only require 2 multiplies
 - o Private key operations remain expensive
 - If $M < N^{1/3}$ then $C = M^e = M^3$ and **cube root attack**
 - For any M , if C_1, C_2, C_3 sent to 3 users, cube root attack works (uses Chinese Remainder Theorem)
- ❑ Can prevent cube root attack by padding message with random bits
- ❑ Note: $e = 2^{16} + 1$ also used ("better" than $e = 3$)

Sample Questions

- Suppose that Alice's RSA public key is $(N, e) = (33, 3)$ and her private key is $d = 7$.
 - a. If Bob encrypts the message $M = 19$ using Alice's public key, what is the ciphertext C ? Show that Alice can decrypt C to obtain M .
- Suppose that for the knapsack cryptosystem, the superincreasing knapsack is $(3, 5, 12, 23)$ with $n = 47$ and $m = 6$.
 - a. Give the public and private keys.
 - b. Encrypt the message $M = 1110$ (given in binary). Give your result in decimal.